08-Conv-exc

Convex sets

1. Låt $A, B \subset \mathbb{R}^n$ vara två konvexa mängder. Visa att mängderna nedan är också konvexa: b4a4c0

   (a) $cA = \{c\mathbf{x} : \mathbf{x} \in A,\ c \in \mathbb{R}\}$ (b) $A + B = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in A,\ \mathbf{y} \in B\}$

2. Låt $f, g \colon \mathbb{R}^n \to \mathbb{R}$ vara konvexa samt $c \geq 0$. Visa att funktionerna nedan är konvexa: 258cf5

   (a) cf (b) $f + g$ (c) $\max\{f, g\}$.

3. Låt $f \colon [a, b] \to \mathbb{R}$ vara convex och växande funktion av en variabel, samt $g \colon \Omega \to [a, b]$ vara convex funktion av flera variabler, $\Omega \subset \mathbb{R}^n$. Visa att den sammansatta funktionen $f(g(\mathbf{x}))$ är en konvex funktion. 73642b

4. Använd triangelolikheten 134f7b

   $$\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{z}\| + \|\mathbf{z} - \mathbf{y}\|$$

   för att visa att varje enhetsklott

   $$B_1(\mathbf{x}_0) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| \leq 1\}$$

   med centrum i $\mathbf{x}_0$ i $\mathbb{R}^n$ är en konvex mängd. (Obs! Beviset fungerar oförändrat i varje normerat rum).

5. Let $\Omega \subset \mathbb{R}^n$ be convex, $a \in \mathbb{R}$, $\mathbf{b} \in \mathbb{R}^n$. Show that the set 8de74a

   $$a\Omega + \mathbf{b} = \{a\mathbf{x} + \mathbf{b},\ \mathbf{x} \in \Omega\},$$

   is also convex (that is, set convexity is scaling and translation invariant).

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6. Let $\Omega \subset \mathbb{R}^n$ be convex, A – a $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Denote by $F \colon \mathbb{R}^n \to \mathbb{R}^m$ be the linear map 8de74b

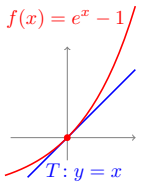
   $$F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}, \quad \text{and by} \quad F(\Omega) = \{\mathbf{y} = F(\mathbf{x}),\ \mathbf{x} \in \Omega\} \subset \mathbb{R}^m \quad \text{– the image of } \Omega \text{ under } F.$$

   Show that $F(\Omega)$ is convex. (Hint: Show that $\mathbf{y}_{1,2} \in F(\Omega) \Rightarrow (1-t)\mathbf{y}_1 + t\mathbf{y}_2 \in F(\Omega)$, $0 < t < 1$).

   (*Moral: Linear maps, e.g., projections, rotations, …, as well as translations preserve convexity.*)

Convex functions via the definition

1. Show that the function $f(x) = e^x - 1$ is strictly convex for all $x \in \mathbb{R}$. Use that the graph of strictly convex functions lies above its tangent at any point to: d2bcb7

   (a) show that $e^x - 1 > x$ for all $x \neq 0$;

   (b) set $x = \dfrac{\pi}{e} - 1$ in (a) to show that $e^\pi > \pi^e$.

2. Let f be convex and strictly increasing function of one variable. Motivate whether its inverse f^{-1} is convex, concave, or, alternatively, nothing can be claimed in that respect. (Reminder: The inverse of an increasing function is, of course, also increasing.) c7fa5d

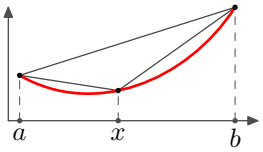
3. To be able to establish convexity without greasing your hands with derivatives is a big saver. 680354

   (a) Let $g : \mathbb{R}^n \to \mathbb{R}$ be convex and $f : \mathbb{R} \to \mathbb{R}$ be concave and decreasing. Is the composition of these functions $f \circ g(\mathbf{x}) \stackrel{\text{def}}{=} f[g(\mathbf{x})]$ convex, concave, neither?

   (b) Let $x^2 + y^2 \leq 1$. Is $\cos(\arcsin(x^2 + y^2))$ convex, concave, neither?

4. Use the definition of a convex function $f \colon \mathbb{R} \to \mathbb{R}$,

   $$f(x) = f[(1-\lambda)a + \lambda b] \leq (1-\lambda)f(a) + \lambda f(b), \quad 0 \leq \lambda = \frac{x-a}{b-a} \leq 1,$$

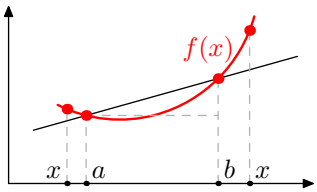
   

   to show that: $\dfrac{f(x) - f(a)}{x - a} \leq \dfrac{f(b) - f(a)}{b - a} \leq \dfrac{f(b) - f(x)}{b - x}$, $a < x < b$.

5. Låt $f:\mathbb{R}\to\mathbb{R}$ vara konvex och strängt växande. Visa att f^{-1} är också strängt växande men konkav. e66ad4
   Tips: Att $f\nearrow$ innebär att $(x_2-x_1)(y_1-y_2)>0$, $y_i=f(x_i)$. (Obs! f behöver inte vara differentierbar).

6. Visa att om $f:\mathbb{R}^n\to\mathbb{R}^1$ är konvex, då för alla $\mathbf{x}$ och $\mathbf{y}$ gäller att: $\displaystyle\int_0^1 f[\mathbf{x}+t(\mathbf{y}-\mathbf{x})]\,dt\le\frac{f(\mathbf{x})+f(\mathbf{y})}{2}$. 8878ab

7. Suppose that $f(x)$ is convex on D_f. Show that for any interval $[a,b]\subset\overset{\circ}{D}_f$, d0e7ce



$$\begin{cases} f(x)\ge(1-t)f(a)+tf(b)\\ t\notin[0,1]\end{cases}\quad\Leftrightarrow\quad\begin{cases} f(x)\ge f(a)+\dfrac{f(b)-f(a)}{b-a}(x-a)\\ x\notin[a,b],\end{cases}$$

   i.e., the graph of f is above the secant via $(a,f(a))$ $(b,f(b))$ for $x\notin[a,b]$, by:

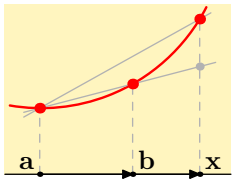
   (a) rewriting the convexity inequality $f(b)\le(1-t)f(a)+tf(x)$, $t=(x-b)/(x-a)$, if $a<b<x$;
   (b) rewriting the convexity inequality $f(a)\le tf(a)+(1-t)f(x)$, $t=(b-a)/(b-x)$, if $x<a<b$.

8. Let $f:\mathbb{R}^n\to\mathbb{R}$ be convex. Show that convexity, $f[(1-t)\mathbf{a}+t\mathbf{b}]\le(1-t)f(\mathbf{a})+tf(\mathbf{b})$, $0\le t\le1$, implies d0e7cf

$$f[(1-t)\mathbf{a}+t\mathbf{b}]\ge(1-t)f(\mathbf{a})+tf(\mathbf{b}),\ t\notin[0,1].$$

   Hint: Let $\mathbf{a},\mathbf{b},\mathbf{x}$ lie in that order on the same line, so that $\mathbf{b}=(1-s)\mathbf{a}+s\mathbf{x}$, $0<s<1$ and use convexity.

9. Show that a convex $f:\mathbb{R}^n\to\mathbb{R}$ cannot be bounded unless it is constant. That is to say, if a convex $f(\mathbf{x})\le M$ for all $\mathbf{x}\in\mathbb{R}^n$, then $f=\text{const}$. cec490
   Hint: Use that if $f\ne\text{const}$ is convex, there exist $\mathbf{a}$ and $\mathbf{b}$ s.t. $f(\mathbf{a})<f(\mathbf{b})$, and



$$\frac{f(\mathbf{b})-f(\mathbf{a})}{\|\mathbf{b}-\mathbf{a}\|}\le\frac{f(\mathbf{x})-f(\mathbf{a})}{\|\mathbf{x}-\mathbf{a}\|}\quad\text{for all}\quad\mathbf{x}=\mathbf{a}+t(\mathbf{b}-\mathbf{a}),\ t\ge1,$$

   i.e., the slope of the secants increases as $\mathbf{x}$ passes beyond $\mathbf{b}$ along the ray from $\mathbf{a}$ through $\mathbf{b}$.

10. For which $\mathbf{x}$ is $f(\mathbf{x})=x_1^3-3x_1^2+x_1x_2+x_2^3-3x_2^2$ a convex function? 67e1f9

Convexity of smooth functions ($\nabla\nabla^T f\succeq0$)

1. Consider the following example of inequality based on the three postive numbers 2, 3 and 5: 1d907f

$$H(2,3,5)=\Big[\frac{1}{3}(\frac{1}{2}+\frac{1}{3}+\frac{1}{5})\Big]^{-1}=\frac{3}{1/2+1/3+1/5}=\frac{90}{31}<3<\frac{10}{3}=\frac{2+3+5}{3}=A(2,3,5).$$

   The left quotient is the harmonic mean of these numbers, the right is their arithmetic mean.

   (a) Prove that the function $f(x)=x^{-1}$ is convex for all $x>0$.
   (b) Use Jensen's inequality to show that the harmonic mean of any collection of n positive numbers cannot exceed their arithmetic mean:

$$H(\mathbf{x})=\frac{n}{x_1^{-1}+\cdots+x_n^{-1}}\quad\le\quad\frac{x_1+\cdots+x_n}{n}=A(\mathbf{x}),\quad\mathbf{x}>\mathbf{0}.$$

2. Consider the following inequality based on the three postive numbers 2, 3 and 5: 1d9080

$$H(x_1,x_2,x_3)=\frac{3}{1/2+1/3+1/5}=\frac{90}{31}<3=\sqrt[3]{27}<\sqrt[3]{(2)(3)(5)}=G(x_1,x_2,x_3).$$

   The left quotient is called the *harmonic* mean of these three numbers, the right being their *geometric* mean.
   Show that the harmonic mean of any n positive numbers cannot exceed their geometric mean:

$$H(\mathbf{x})\overset{\text{def}}{=}\frac{n}{x_1^{-1}+\cdots+x_n^{-1}}\quad\le\quad(x_1\cdots x_n)^{1/n}\overset{\text{def}}{=}G(\mathbf{x}),\quad\mathbf{x}>\mathbf{0}.$$

   Hint: Consider $\ln H(\mathbf{x})$ and use the Jensen's inequality on the convex function $f(x)=\ln(x^{-1})=-\ln(x)$.

3. En reel $n \times n$ matris A kallas positiv definit ($A \succ 0$) om den skalära produkten (kvadratiska formen) 43bdf8

$$q(\mathbf{x}) = \mathbf{x} \circ A\mathbf{x} = \mathbf{x}^T A\mathbf{x} > 0, \ \forall \mathbf{x} \in \mathbb{R}^n,$$

dvs, $\mathbf{x}$ och $A\mathbf{x}$ slutar spetsig vinkel. Visa att om $A = A^T$, då är alla dess egenvärden icke-negativa.

Tips: Enligt Spektralsatsen, symmetriska matriser har reella egenvärden och är ortogonalt diagonaliserbara:

$$q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} = \mathbf{x}^T (P\Lambda P^T)\mathbf{x}, \quad \text{där} \quad \Lambda = \operatorname{diag}(\lambda_1, \cdots, \lambda_n), \quad \text{och} \quad P^{-1} = P^T.$$

4. Låt $A = B^T B$, där B är en godtycklig (inte nödvändigtvis kvadratisk) matris. Visa att $A \succeq 0$ (är positiv definit), dvs, den skalära produkten $\mathbf{x} \circ A\mathbf{x} \geq 0$, $\forall \mathbf{x}$. 5f361c

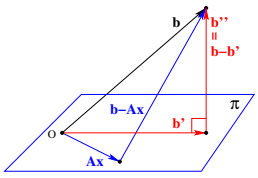
5. En reel $n \times n$ matris A ger upphov till en kvadratisk form $q\colon \mathbb{R}^n \to \mathbb{R}$ genom den skalära produkten 945bf0

$$q(\mathbf{x}) = \mathbf{x} \circ A\mathbf{x} = \mathbf{x}^T A\mathbf{x}, \ \mathbf{x} \in \mathbb{R}^n.$$

Visa att $q(\mathbf{x}) = \mathbf{x}^T B\mathbf{x}$, där $B = \dfrac{A + A^T}{2} = B^T$.

Convex problems

1. (Least Squares as a convex min problem). Let A be a real $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. If $m > n$, the system $A\mathbf{x} = \mathbf{b}$ is generally unsolvable (unless, of course, $\mathbf{b} \in \pi : \operatorname{span}\operatorname{cols}(A)$). One can define a *best solution*, in the sense it minimizes the squared length of the error, d76bda



$$\min f(\mathbf{x}) = \|\mathbf{b} - A\mathbf{x}\|^2, \quad \mathbf{x} \in \mathbb{R}^n, \qquad (\star)$$

   (a) Show that the problem ($\star$) is convex, so its local min $\mathbf{x}^*$ is also global. Find $\mathbf{x}^*$! (Hint: For any square matrix $B_{n\times n}$ and $\mathbf{x} \in \mathbb{R}^n$, the gradient $\nabla(\mathbf{x}^T B\mathbf{x}) = \nabla(\mathbf{x}^T B^T\mathbf{x}) = (B^T + B)\mathbf{x}$.)

   (b) Find A and $\mathbf{b}$ to compose a problem ($\star$), the solution $\mathbf{x}^* = (a, b, c)^T$ of which finds the polynomial $p_2(t) = a + bt + ct^2$ best fitting the data to the right. (**Optional**) Find that best fit p_2!

t	-1	0	1	2
$p_2(t)$	5	6	-1	4

2. Motivate whether the constrained optimization problem below is convex: cea56b

$$\max f(\mathbf{x}) = 2x_1 - 3x_2 + \ln(5 + x_1 + x_2) - 3e^{-x_1 + 2x_2^2}$$
$$\textbf{s.t.} \begin{cases} x_1^2 - 2x_1 + x_2^2 & \leq 4 \\ x_1^2 + x_2^2 - 2x_2 & \leq 4 \\ x_1 - x_2 & \leq 0 \end{cases}$$

Parallelized

1. Utnyttja att bilden $F(\Omega)$ av en konvex mängd $\Omega \subset \mathbb{R}^n$ under avbildningen $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ är konvex, där A är matris och $\mathbf{b} \in \mathbb{R}^n$, för att visa att ellipsen 81fd47

$$E\colon \frac{(y_1 + 4)^2}{2^2} + \frac{(y_2 - 6)^2}{3^2} + \frac{(y_3 - 4)^2}{4^2} \leq 1$$

är konvex. Tips: $E = F(\text{enhetsklottet})$ under viken avbidning $\mathbf{y} = F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, $\|\mathbf{x}\| \leq 1$?

2. If $f(x) = x \ln x$, $x > 0$, show that $f''(x) > 0$. Hence f is a convex function. Let now $\mathbf{p} = \{p_i\}_1^n$ be any discrete distribution on $0 < x_1 < \cdots < x_n$, with $p_i = \text{Prob}(X = x_i)$. Use Jensen's inequality to show that 317974

$$f(p_1 x_1 + \cdots + p_n x_n) = f(\mu_X) = f[E(X)] \leq E[f(X)] = \mu_{f(x)} = p_1 f(x_1) + \cdots + p_n f(x_n).$$

As an illustration, check that $E(X) \leq E[f(X)]$ for the expected values of the distribution below:

x_i	2	6	8	11	12
p_i	0.21	0.29	0.15	0.22	0.13